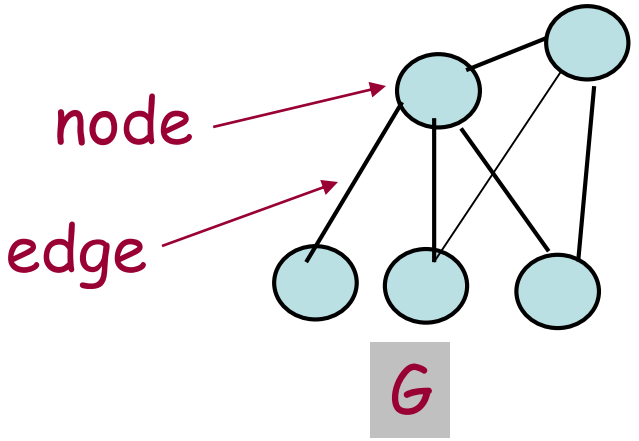
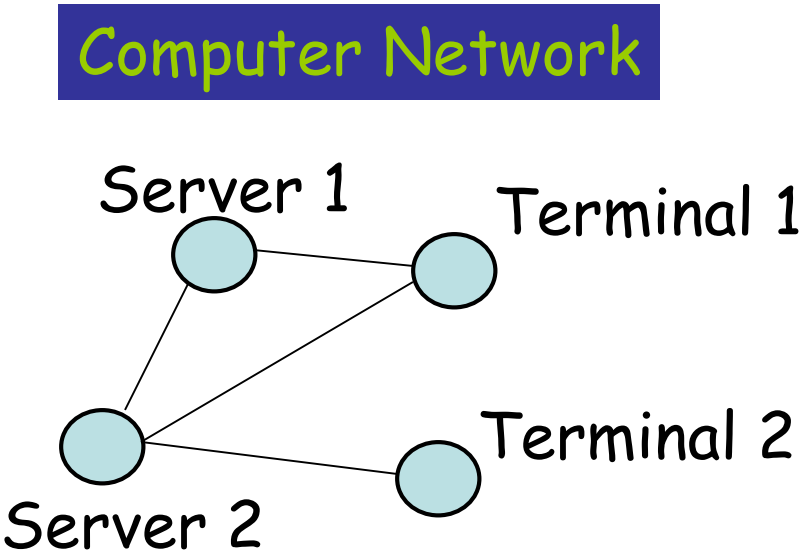
Graphs

What are graphs?

- Graphs represent the relationships among data items
- A simple graph G consists of
  - a set V of nodes (vertices)
  - a set E of edges: each edge connects two nodes
- For any nodes u and v, if u and v are connected by an edge, such edge is denoted as (u, v)



Example



Other examples: electrical and communication networks, airline routes, WWW, etc.

Applications of Graphs

- Potentially anything (graphs can represent relations, relations can describe the extension of any predicate).

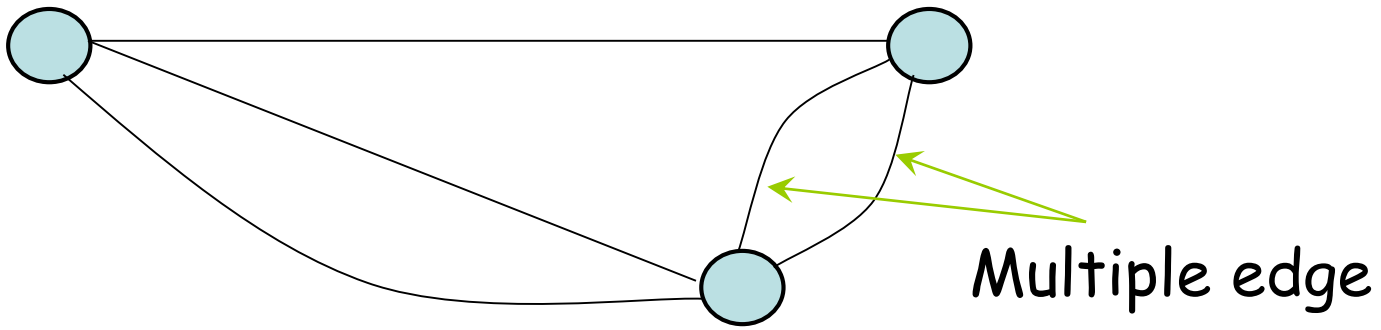
- Applications in networking, scheduling, optimization, circuit design, path planning.

Multigraph

Multigraph allows multiple edges (self-loop not allowed).

- A multigraph $G = (V, E, f)$ consists of a set V of vertices, a set E of edges (as primitive objects), and a function $f: E \rightarrow \{\{u,v\} | u,v \in V \wedge u \neq v\}$.

- E.g., nodes are cities, edges are segments of major highways.

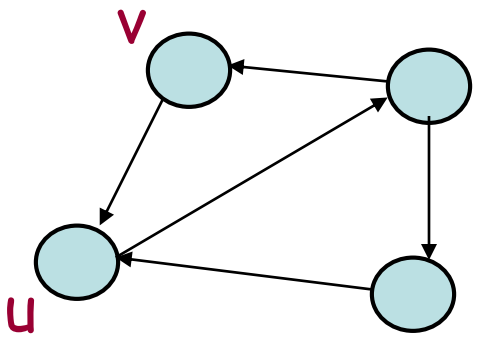


Directed Graph

A directed graph G = (V, E) consists of a set V of vertices and a set E of edges that are ordered pairs of elements in V.

In otherwords, a directed graph (V,E) consists of a set of vertices V and a binary relation E on V.

Example: V = set of People; E={(x,y) | x loves y}



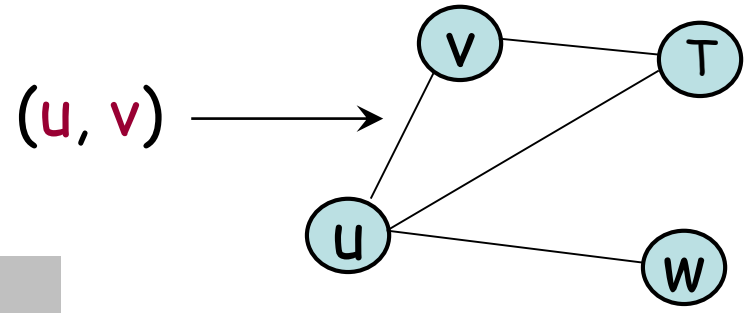
Graph Terminology - Adjacency

Two vertices u and v in an undirected graph G are called adjacent (or neighbors) in G if (u, v) is an edge in G.

- If e = (u, v), the edge e is called incident with the vertices u and v.

- The vertices u and v are called endpoints of the edge (u,v).

Example



u and v are adjacent
v and w are not adjacent

Graph Terminology - Adjacency

When (u, v) is an edge of a directed graph G, u is said to be adjacent **to** v, and v is said to be adjacent **from** u.

- The vertex u is called the initial vertex of (u, v), and v is called the terminal vertex of (u, v).

- The initial vertex and terminal vertex of a loop are the same.

Graph Terminology

The degree of a vertex in an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex.

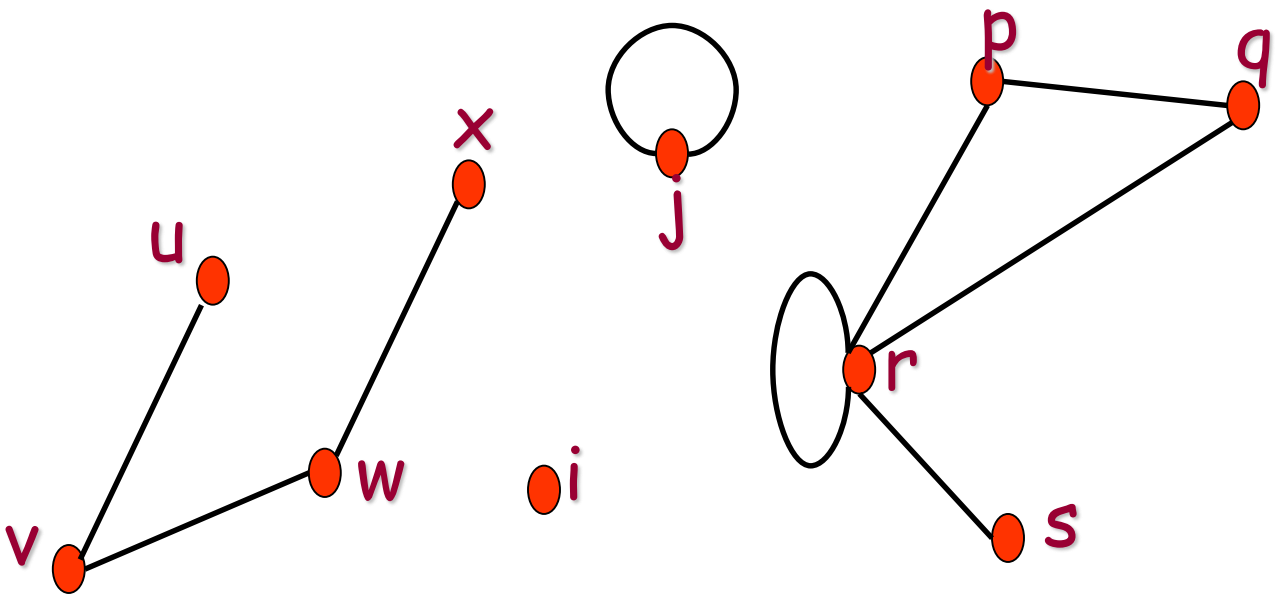
- The degree of the vertex v is denoted by deg(v).

•A vertex of degree 0 is called isolated, since it is not adjacent to any vertex.

•A vertex with a loop at it has at least degree 2 and it is not isolated, even if it is not adjacent to any other vertex.

•A vertex of degree 1 is called pendant. It is adjacent to exactly one other vertex.

Example



- Vertex i is isolated
- vertices u, x and s are pendant.
- degree of r, deg(r) = 5.

The Handshaking Theorem: Let $G = (V, E)$ be an undirected graph with e edges. Then $2e = \sum_{v \in V} \deg(v)$.

Theorem: An undirected graph has an even number of vertices of odd degree.

Proof: Let V_1 and V_2 be the set of vertices of even and odd degrees, respectively (Thus $V_1 \cap V_2 = \varnothing$, and $V_1 \cup V_2 = V$).

Then by Handshaking theorem

$$2|E| = \sum_{v \in V} \deg(v) = \sum_{v \in V1} \deg(v) + \sum_{v \in V2} \deg(v)$$

Since both $2|E|$ and $\sum_{v \in V1} \deg(v)$ are even, $\sum_{v \in V2} \deg(v)$ must be even.

As $\deg(v)$ is odd, $\forall v \in V_2$, $|V_2|$ must be even.

In-degrees and Out-degrees

In a directed graph, the in-degree of a vertex v, denoted by $\deg^-(v)$, is the number of edges with v as their terminal vertex.

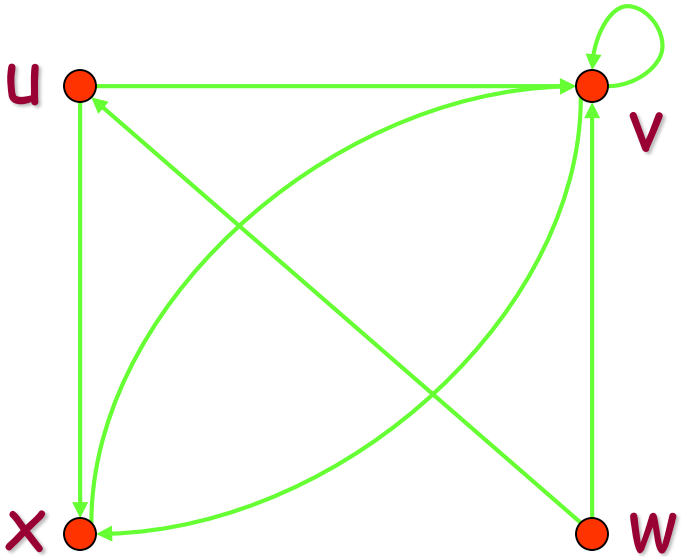
The out-degree of v, denoted by $\deg^+(v)$, is the number of edges with v as their initial vertex.

Note: A loop to a vertex increases both the in-degree and the out-degree by one.

Example

What are the in-degrees and out-degrees of the vertices u, v, w, x in this graph ?

- $\deg^-(u) = 1$
- $\deg^+(u) = 2$



- $\deg^-(v) = 4$
- $\deg^+(v) = 2$

- $\deg^-(x) = 2$
- $\deg^+(x) = 1$

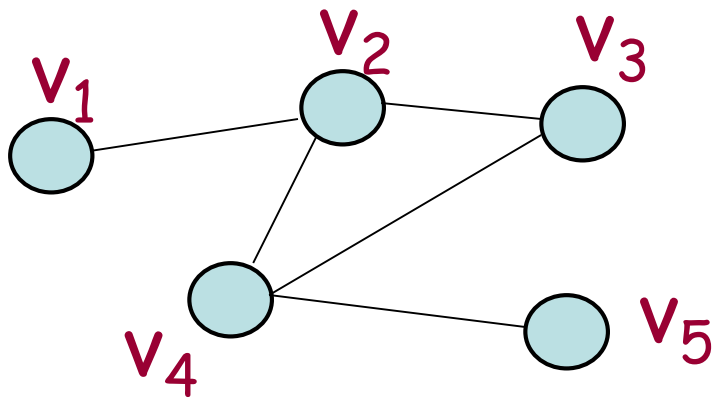
- $\deg^-(w) = 0$
- $\deg^+(w) = 2$

Theorem: Let $G = (V, E)$ be a graph with directed edges. Then $\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|$

Path and simple path

- A path of length n from u to v, where n is a positive integer, in an undirected graph is a sequence of edges e_1, e_2,…, e_n of the graph such that $e_1 = (v_0, v_1)$, $e_2 = (v_1, v_2)$,…, $e_n = (v_{n-1}, v_n)$, where $v_0 = u$ and $v_n = v$.
- A path is called a simple path if every node appears at most once.

- v_2, v_3, v_4, v_2, v_1 is a path
- v_2, v_3, v_4, v_5 is a simple path



Walk

Let u and v be two vertices in a graph G. A **walk** from u to v, in G, is an alternating sequence of $n + 1$ vertices and n edges of G

$$(u = v_1, e_1, v_2, e_2, v_3, e_3, \ldots, v_{n-1}, e_{n-1}, v_n, e_n, v_{n+1} = v)$$

beginning with vertex u, called the **initial vertex**, and ending with vertex v, called the **terminal vertex**, in which v_i and v_{i+1} are endpoints of edge e_i for $i = 1, 2, \ldots, n$.

Note

A walk with no repeated edges is called a **trail**, and a walk with no repeated vertices except possibly the initial and terminal vertices is called a **path**.

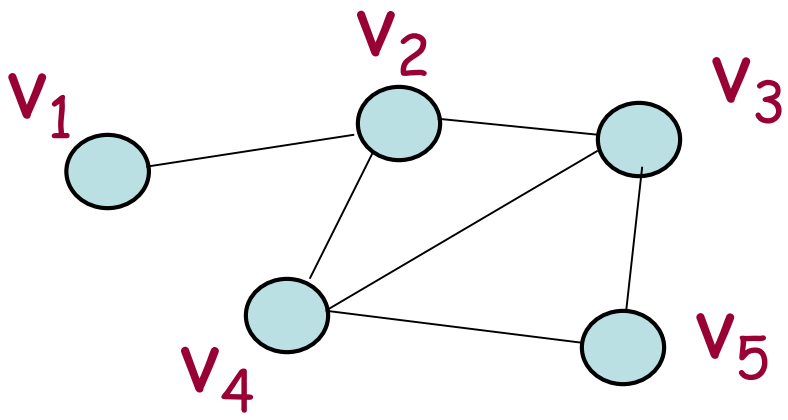
Cycle and simple cycle

A cycle of n vertices $v_1, v_2, \ldots, v_n$, and edges $\{v_1, v_2\}, \{v_2, v_3\}, \ldots, \{v_{n-1}, v_n\}, \{v_n, v_1\}$, denoted by C_n, $n \geq 3$.

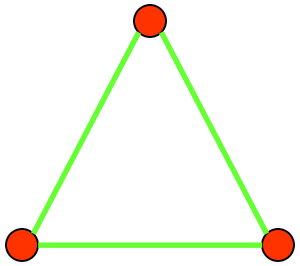
- A cycle is a path that begins and ends at the same vertex.

- A cycle is a simple cycle if every vertex appears at most once, except for the first and the last vertices.

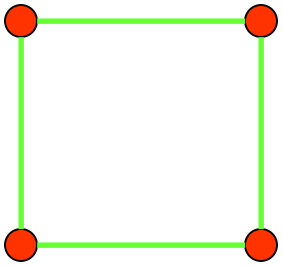
- $v_2, v_3, v_4, v_5, v_3, v_2$ is a cycle
- v_2, v_3, v_4, v_2 is a simple cycle



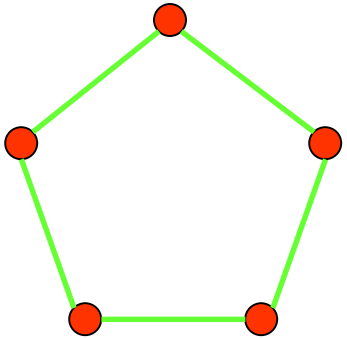
More Examples



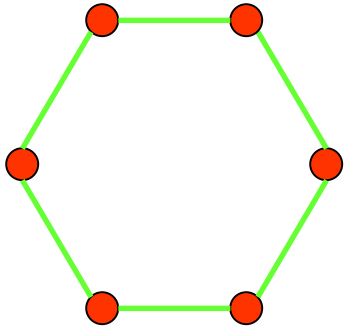
C_3



C_4



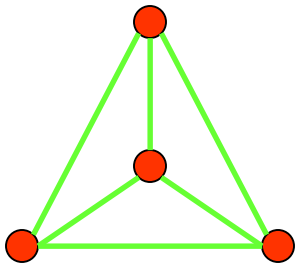
C_5



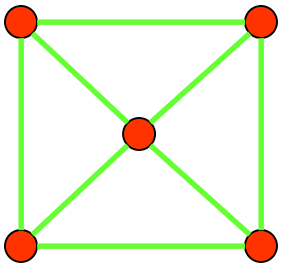
C_6

Wheel

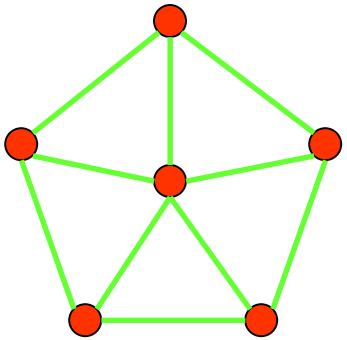
The wheel, W_n, can be constructed by adding an additional vertex to the cycle C_n, for $n \geq 3$, and connecting the new vertex to each of the n vertices in C_n by adding new edges.



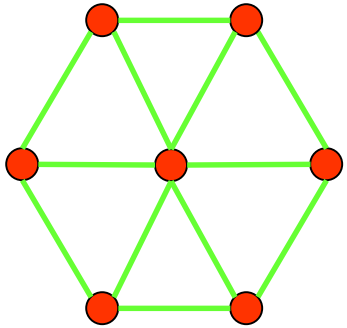
W_3



W_4



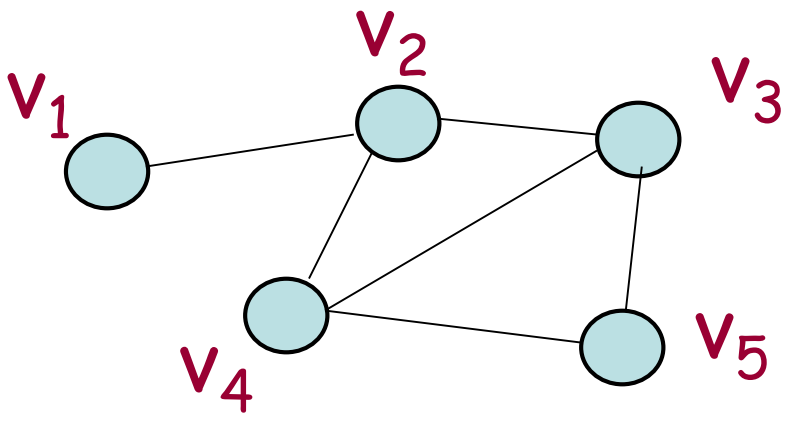
W_5



W_6

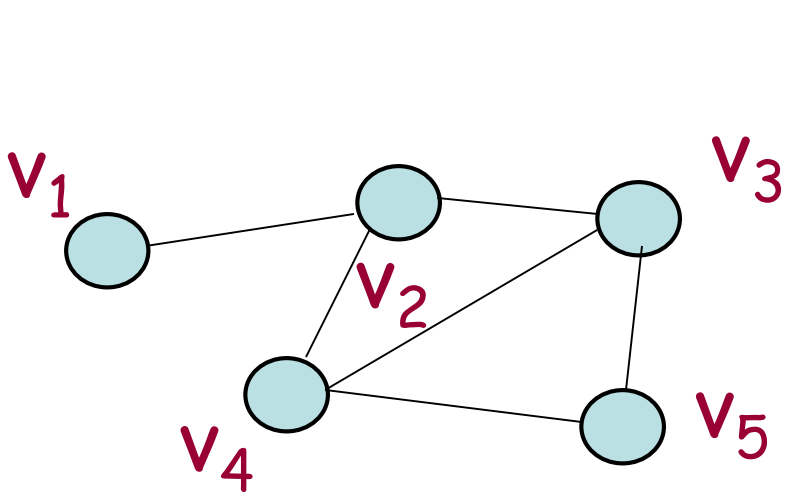
Connected graph

A graph G is connected if there exists path between every pair of distinct vertices.

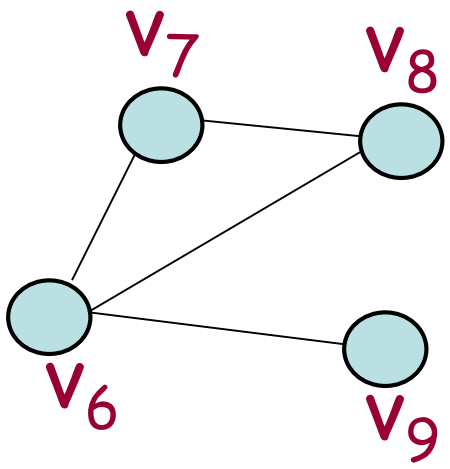


Note: A graph consisting of only one vertex is always connected, because it does not contain any pair of distinct vertices.

Example of disconnected graph



G

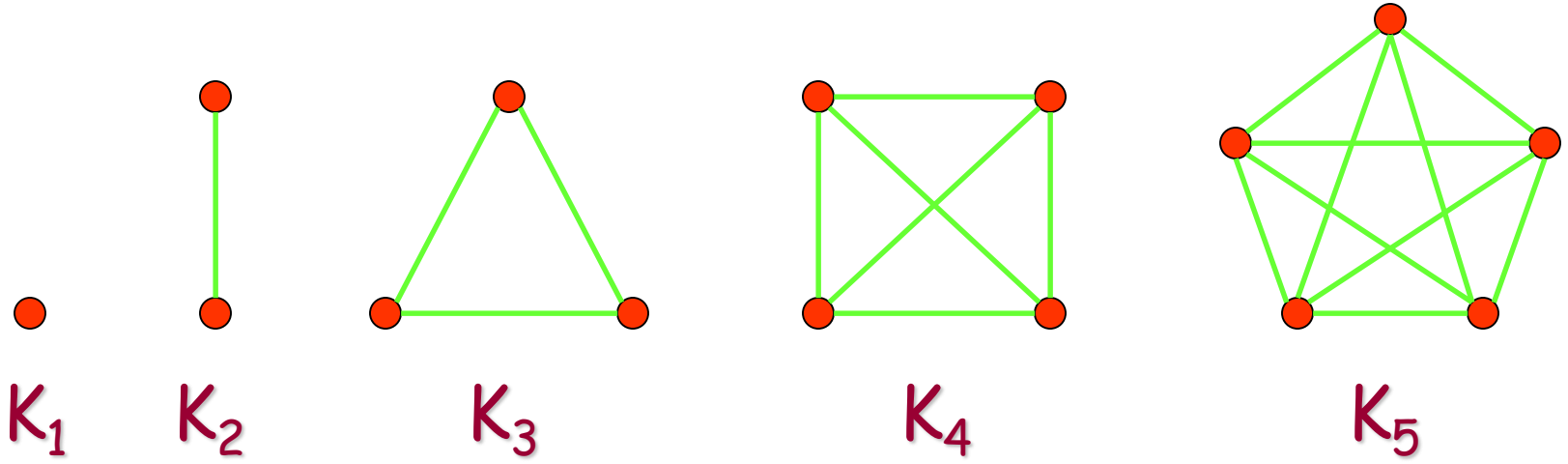


H

G and H are disconnected

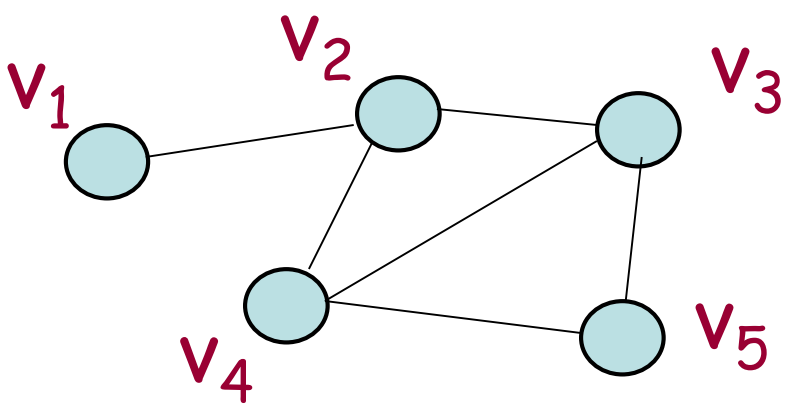
Complete graph

The complete graph on n vertices, denoted by K_n, is the simple graph that contains exactly one edge between each pair of distinct vertices.

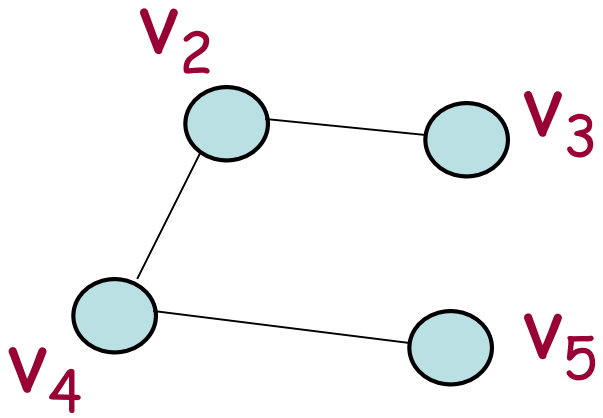


Subgraph

A subgraph of a graph $G = (V, E)$ is a graph $H = (U, F)$ such that $U \subseteq V$ and $F \subseteq E$.



G



H

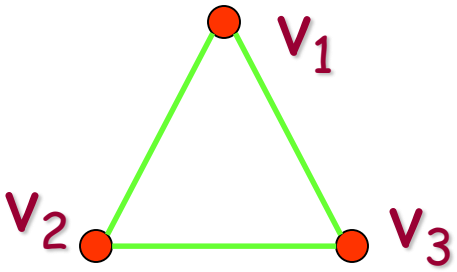
Bipartite Graphs

A simple graph is called bipartite if its vertex set V can be partitioned into two disjoint nonempty sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 with a vertex in V_2 (no edge in G connects either two vertices in V_1 or two vertices in V_2).

Example: Job matching problem. Suppose we have a set of people P and a set of jobs J, with not all people suitable for all jobs. We can model this as a graph with $P + J$ the set of vertices. If a person p_i is suitable for a certain job j_i there is an edge between p_i and j_i in the graph.

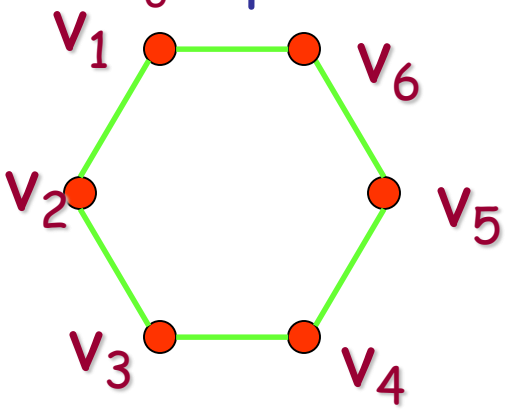
Examples

Is C_3 bipartite?

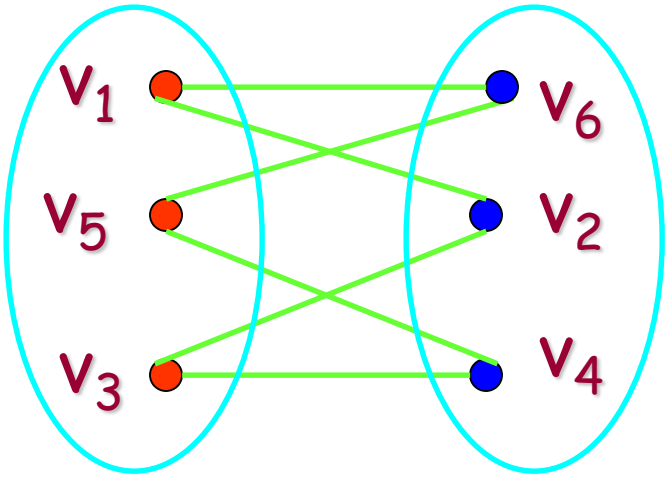


- No, because there is no way to partition the vertices into two sets so that there are no edges with two vertices in the same set.

Is C_6 bipartite?



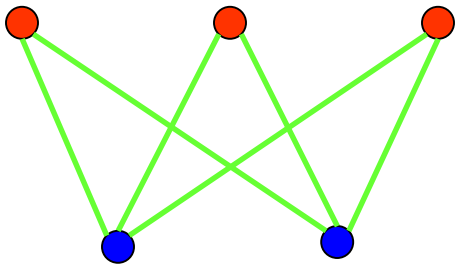
- Yes, because we can partition C_6 and it follows the properties of bipartite graph.



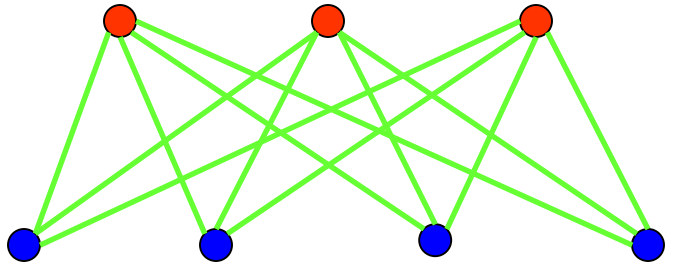
Note: A graph is bipartite if and only if it does not contain any cycle of odd length.

Complete Bipartite graph

The complete bipartite graph $K_{m,n}$ is the graph that has its vertex set partitioned into two subsets of m and n vertices, respectively. Two vertices are connected if and only if they are in different subsets.



$K_{3,2}$



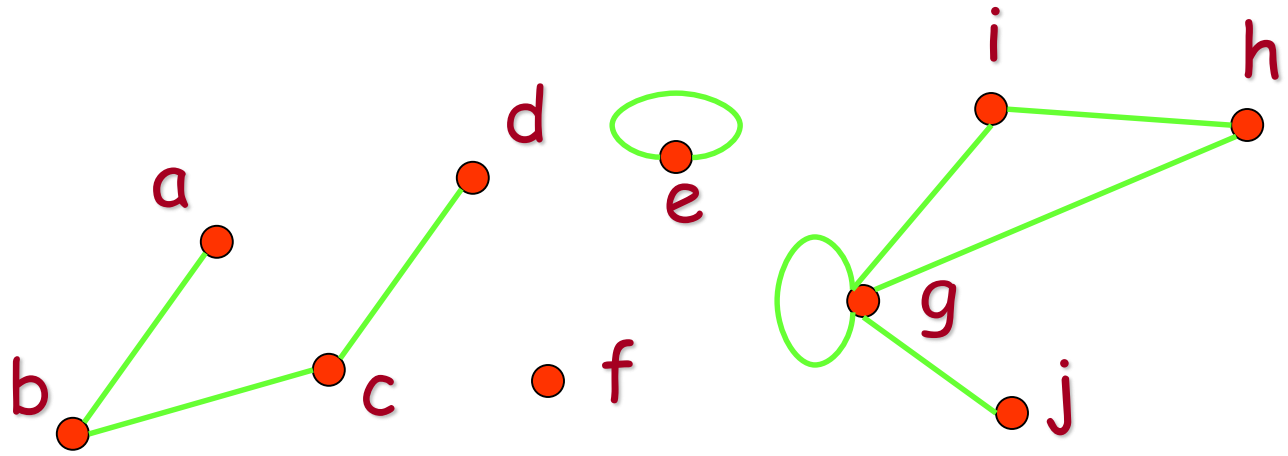
$K_{3,4}$

Connectivity

A graph that is not connected is the union of two or more connected subgraphs, each pair of which has no vertex in common. These disjoint connected subgraphs are called the connected components of the graph.

Connectivity

What are the connected components in the following graph?



Solution: The connected components are the graphs with vertices {a, b, c, d}, {e}, {f}, {i, h, g, j}.

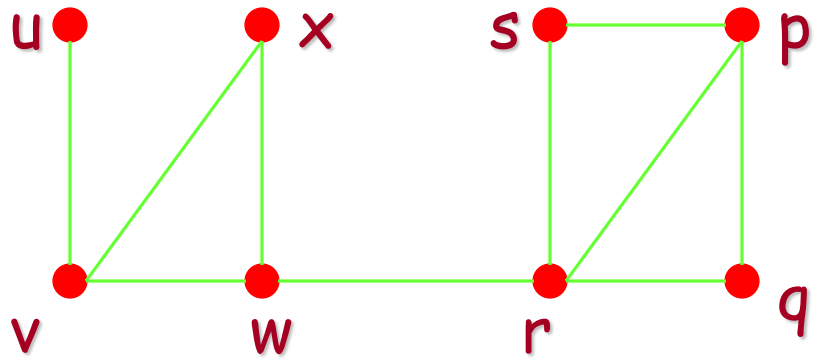
Cut Vertex and Cut Edge

Sometime on removal of a vertex and all edges incident with it produces a subgraph with more connected component than in the original graph. Such a vertex is called cut vertex.

An edge whose removal produces a graph with more connected component than in the original graph is called cut edge.

Example

Find the cut vertices and cut edges of this graph.



cut vertices are v, w, r.

cut edges are {u, v}, {w, r}.

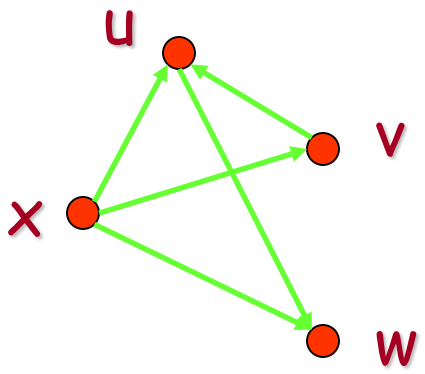
Strongly/Weakly connected

A directed graph is strongly connected if there is a path from u to v and from v to u whenever u and v are vertices in the graph.

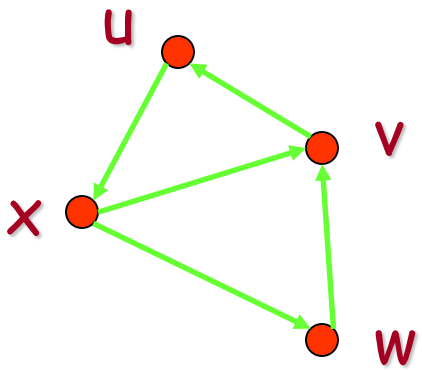
A directed graph is weakly connected if there is a path between every two vertices in the underlying undirected graph.

Example

Are the following directed graphs strongly or weakly connected ?



- Weakly connected, because, for example, there is no path from v to x.



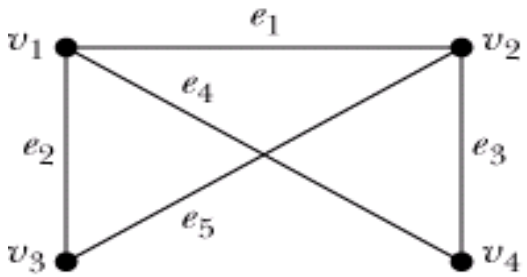
- Strongly connected, because there are paths between all possible pairs of vertices.

Isomorphism of Graphs

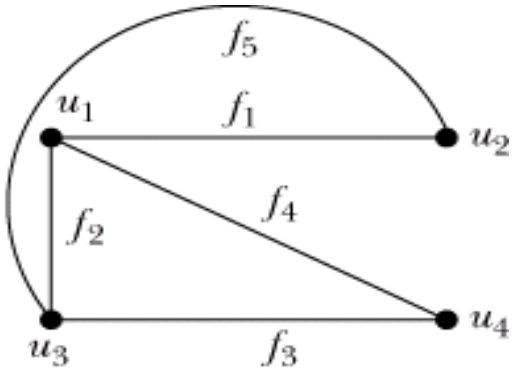
The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there is a bijection (one-to-one and onto function) f from V_1 to V_2 with the property that u and v are adjacent in G_1 if and only if $f(u)$ and $f(v)$ are adjacent in G_2, for all u and v in V_1. Such a function f is called an isomorphism.

In other words, G_1 and G_2 are isomorphic if their vertices can be ordered in such a way that the adjacency matrices A_{G_1} and A_{G_2} are identical.

Isomorphism of Graphs: example



G_1



G_2

- v_1 and v_2 are the end vertices of edge e_1 in G_1 and $f(v_1) = u_1$ and $f(v_2) = u_3$ are adjacent in G_2. Similarly, check for other vertices.

- G_1 is isomorphic to G_2

Isomorphism of Graphs

For this purpose we can check invariants, that is, properties that two isomorphic simple graphs must have. These are:

- the same number of vertices,
- the same number of edges, and
- the same degrees of vertices.

Note: Two graphs that differ in any of these invariants are not isomorphic, but two graphs that match in all of them are not necessarily isomorphic.

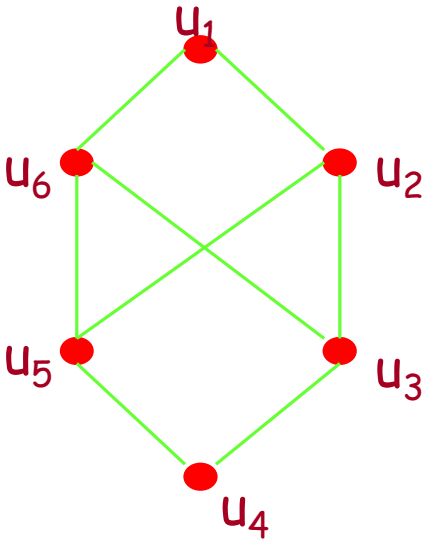
Paths/Circuits and Isomorphism

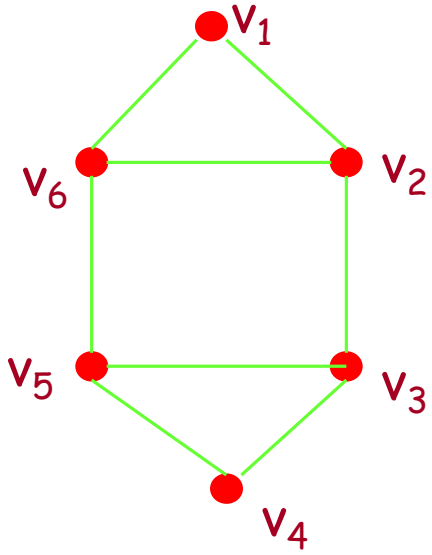
There exists an useful invariant based on the length of a simple circuit and it can be used to show that two graphs are not isomorphic.

Isomorphic Invariant: Two graphs must have simple circuit of same length k (>2).

Paths/Circuits and Isomorphism

Check the isomorphism of the following graphs.

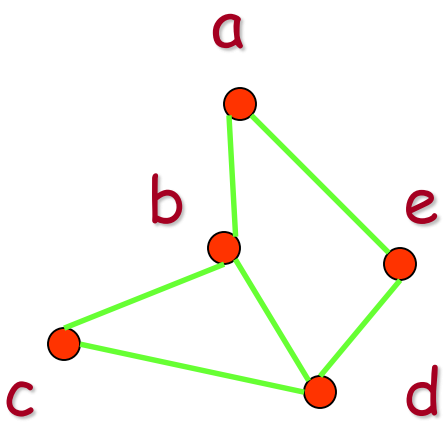


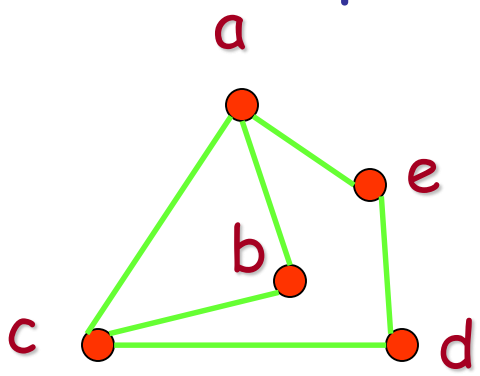


Solution: No. Since the left graph has simple circuit of length at least 4 where as the right graph has simple circuit of length 3.

Isomorphism of Graphs

Are the following two graphs isomorphic?





Yes, they are isomorphic, because they can be arranged to look identical. You can see this if in the right graph you move vertex b to the left of the edge {a, c} and the left graph is rotated almost 90°.

Isomorphism of Graphs



Now the isomorphism f from the left to the right graph is:

f(a) = e,

f(b) = a,

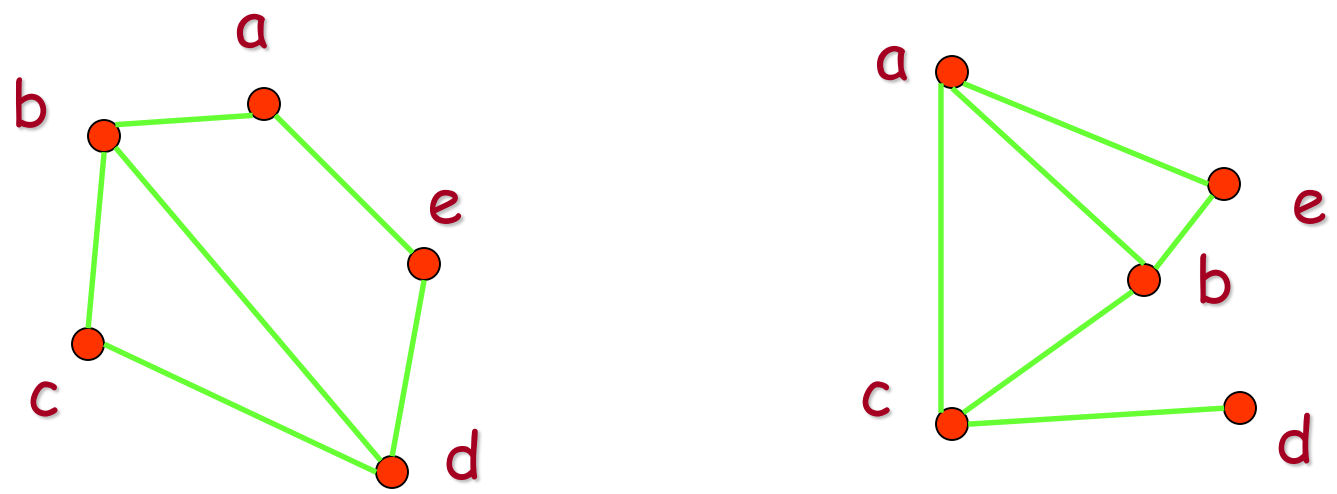
f(c) = b,

f(d) = c,

f(e) = d.

Isomorphism of Graphs

Are the following two graphs isomorphic?



No, they are not isomorphic, because they differ in the degrees of their vertices.

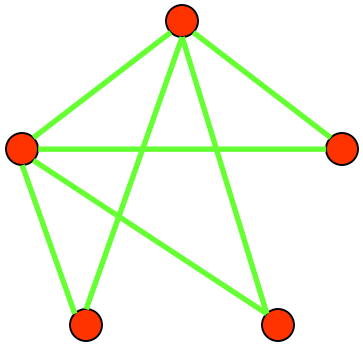
Vertex d in right graph is of degree one, but there is no such vertex in the left graph.

GMT, Combinatorica, are the toolkits to test isomorphisms of graphs.

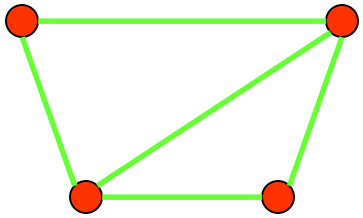
Operations on Graphs - Union

The union of two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is the simple graph with vertex set $V_1 \cup V_2$ and edge set $E_1 \cup E_2$.

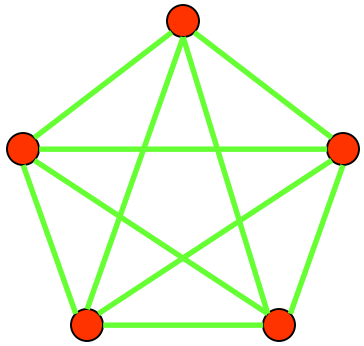
The union of G_1 and G_2 is denoted by $G_1 \cup G_2$.



G_1



G_2

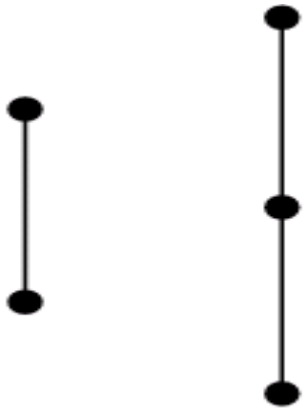


$G_1 \cup G_2$

The union $G_1 \cup G_2$ of two simple graphs $G_1=(V_1, E_1)$ and $G_2=(V_2,E_2)$ is the simple graph $(V_1 \cup V_2, E_1 \cup E_2)$.

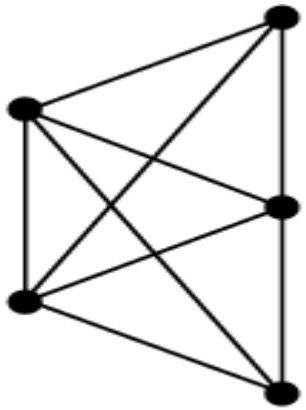
Operations on Graphs - Join

The Join of two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, is the graph union $G_1 \cup G_2$ together with all the edges joining V_1 and V_2.



G_1

G_2



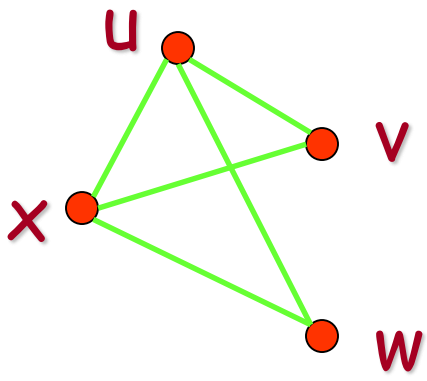
Join(G_1 , G_2)

Graph Representations

Graph representations:

- Adjacency lists.
- Adjacency matrices.
- Incidence matrices.

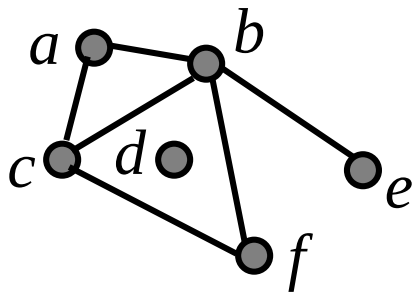
Representing Graphs



Vertex	Adjacent Vertices
u	v, w, x
v	u, x
w	u, x
x	u, v, w

Adjacency Lists

A table with 1 row per vertex, listing its adjacent vertices.



Vertex	Adjacent Vertices
a	*b, c*
b	*a, c, e, f*
c	*a, b, f*
d	
e	*b*
f	*c, b*

Adjacency Matrix

Let $G = (V, E)$ be a simple graph with $|V| = n$.

The adjacency matrix A (or A_G) of G, with respect to the ordering of the vertices, is the $n \times n$ zero-one matrix with 1 as its ij-th entry when v_i and v_j are adjacent, and 0 otherwise.

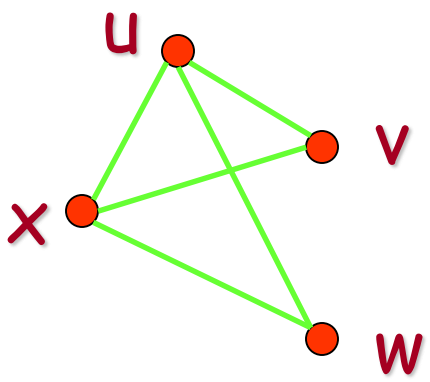
In other words, for an adjacency matrix $A = [a_{ij}]$,

$a_{ij} = 1$ if $(v_i, v_j) \in E$,

$= 0$ otherwise.

Adjacency Matrix: Example

What is the adjacency matrix A_G for the following graph G based on the order of vertices u, v, w, x ?



$$A_G = \begin{array}{c} \\ u \\ v \\ w \\ x \end{array} \begin{array}{c} \begin{array}{cccc} u & v & w & x \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

Adjacency matrices of undirected graphs are always symmetric.

Incidence Matrix

Let $G = (V, E)$ be an undirected graph with $|V| = n$. Suppose $v_1, v_2, \ldots, v_n$ are the vertices and $e_1, e_2, \ldots, e_m$ are the edges of G.

- The incidence matrix of G with respect to this ordering of V and E is the $n \times m$ zero-one matrix with 1 as its ij-th entry when edge e_j is incident with v_i, and 0 otherwise.

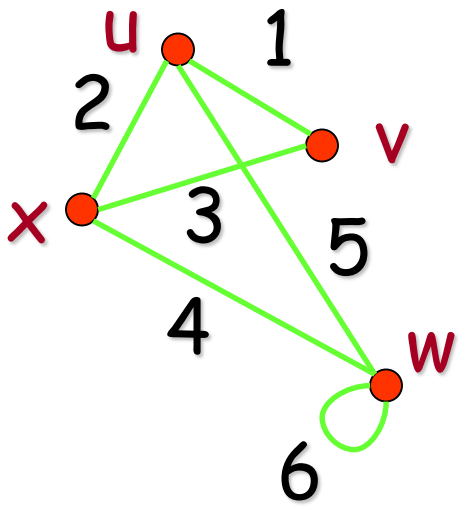
- In other words, for an incidence matrix $M = [m_{ij}]$,

$m_{ij} = 1$ if edge e_j is incident with v_i

$= 0$ otherwise.

Incidence Matrix: Example

What is the incidence matrix M for the following graph G based on the order of vertices u, v, w, x and edges 1, 2, 3, 4, 5, 6 ?



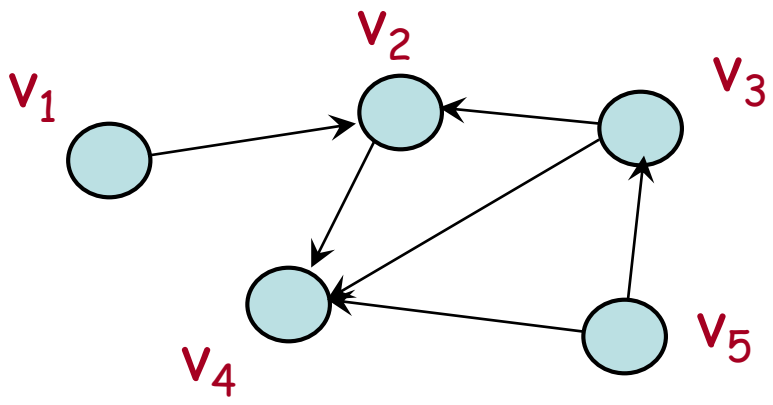
$$M = \begin{array}{c} \\ u \\ v \\ w \\ x \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

Implementing Graph

- Adjacency matrix
  - Represent a graph using a two-dimensional array

- Adjacency list
  - Represent a graph using n linked lists where n is the number of vertices

Adjacency matrix : directed graph



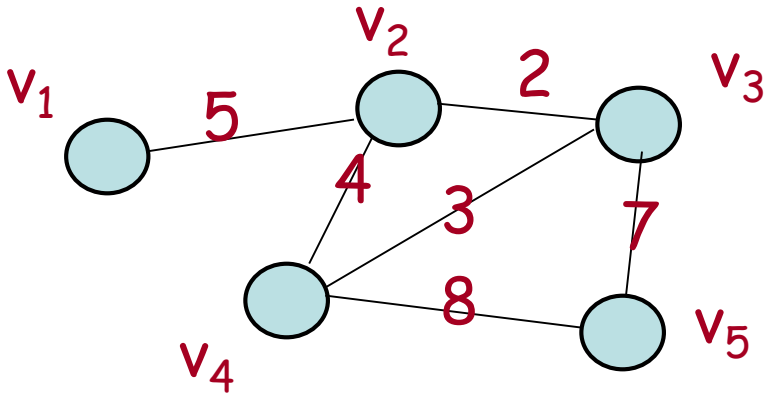
G

$A[i][j] = 1 \quad \text{if } (v_i, v_j) \in E$

$= 0 \quad \text{if } (v_i, v_j) \notin E$

	v_1	v_2	v_3	v_4	v_5
v_1	0	1	0	0	0
v_2	0	0	0	1	0
v_3	0	1	0	1	0
v_4	0	0	0	0	0
v_5	0	0	1	1	0

Adjacency matrix: weighted undirected graph



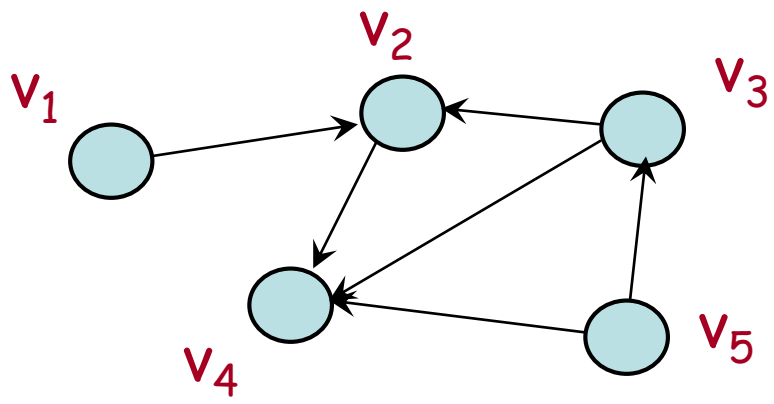
G

	v_1	v_2	v_3	v_4	v_5
v_1	∞	5	∞	∞	∞
v_2	5	∞	2	4	∞
v_3	∞	2	∞	3	7
v_4	∞	4	3	∞	8
v_5	∞	∞	7	8	∞

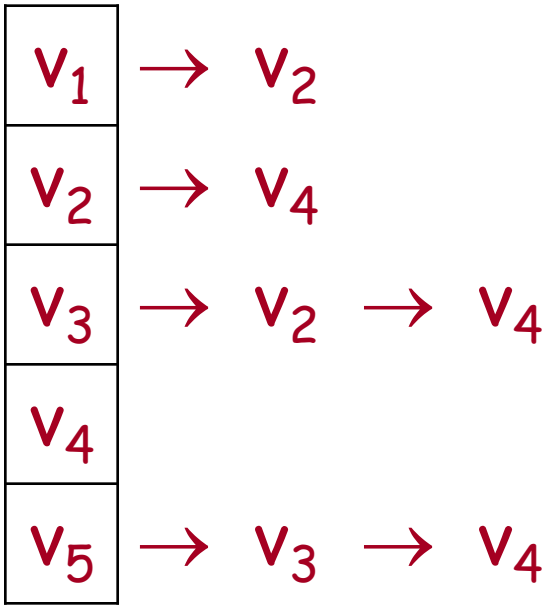
$A[i][j] = w\{v_i, v_j\}$ if $(v_i, v_j) \in E$ or $(v_j, v_i) \in E$

$= \infty$ otherwise

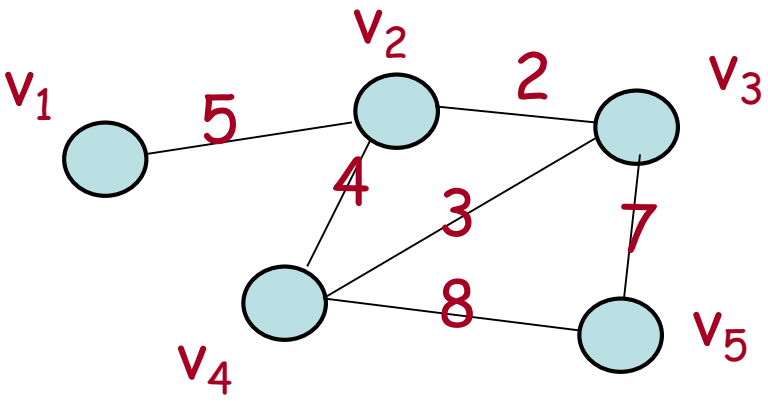
Adjacency list: directed graph



G



Adjacency list: weighted undirected graph



G

v_1	→	$v_2(5)$						
v_2	→	$v_1(5)$	→	$v_3(2)$	→	$v_4(4)$		
v_3	→	$v_2(2)$	→	$v_4(3)$	→	$v_5(7)$		
v_4	→	$v_2(4)$	→	$v_3(3)$	→	$v_5(8)$		
v_5	→	$v_3(7)$	→	$v_4(8)$				

Merits and Limitations

- Adjacency matrix
  - Allows us to determine whether there is an edge from node i to node j efficiently.
  - Storage overhead.

- Adjacency list
  - Allows us to find all nodes adjacent to a given node j efficiently.
  - If the graph is sparse, adjacency list requires less space.